For what values of λ and μ the following system of linear equation has

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

(i) No sol^n

(ii) more than one solution

(iii) unique solution.

$\underline{sol^n:}$ The Augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \xrightarrow[r_3' = -r_1 + r_3]{r_2' = -r_1 + r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right]$$

$$\begin{array}{l} r_1' = -r_2 + r_1 \\ r_3' = -r_2 + r_3 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

$$x - z = 2$$
$$y + 2z = 4$$
$$(\lambda-3) z = \mu - 10 \quad \text{—— (iii)}$$

(i) when $\lambda = 3$, (iii) $\Rightarrow 0 =$ any values of μ (except $\mu \neq 10$)
$\mu \neq 10$, we will have
have no solution.

(ii) when $\lambda \neq 3$ and μ may have any value, a unique solⁿ.

(iii) when $\lambda = 3$ and $\mu = 10$, we will have more than one solution.